Elixir has had fairly large advances in deployment patterns over the last few years. In 2019,[1] the concept of Mix releases was made available. We'll walk through the benefits of releases and how you can use them. After that, we'll cover a few options for observing your app.

Deploy an Elixir App

You could bundle up your application and run it in production using `mix phx.server`. But there's a better option that will reduce long-term friction—Mix releases.[2]

A release is a self-contained package that bundles your compiled code alongside the Erlang VM and runtime. It provides benefits such as code preloading, more configuration options, and not requiring source code in production. It also includes management scripts that make it possible to do things like connect to a running application.

Phoenix has a deployment guide[3] that walks you through deploying an application using Mix releases. We won't walk through the guide here, but this is the best way to deploy your first app. It even includes a Dockerfile that you can use to spin up a containerized version of your application on AWS, GCP, Fly.io, Render, or other cloud providers.

Once you have your app running in production, you need to know if it's working properly.

Observe Your Production Apps

Observability is the art of monitoring your application so that you know what's good and bad with it. There are many different types of observability. Application logging, bug reporting, backend performance monitoring, and frontend tracking are all examples of observability.

The Telemetry[4] library serves as the modern foundation for collecting observability metrics. Telemetry allows libraries like Phoenix, Ecto, and Oban to provide you with metrics about performance. Each of these libraries emits Telemetry events that can be consumed by your Telemetry handlers. The entire system is standardized, so your handlers and a library's events work on top of the same foundation.

1. https://elixir-lang.org/blog/2019/06/24/elixir-v1-9-0-released/
2. https://hexdocs.pm/mix/Mix.Tasks.Release.html
3. https://hexdocs.pm/phoenix/releases.html
4. https://hexdocs.pm/telemetry/readme.html